

Go Data Profiler — Master Technical Copy

Architecture, implementation, operation, analysis, benchmarking, and future evolution.

[illegible][illegible][illegible]

1. Executive Summary

[illegible]

2. Why This Tool Was Built

[illegible][illegible]

[illegible]

10. Basic API Example

[illegible]

```
curl -X POST http://localhost:8080/api/v1/profile-jobs -H 'Content-Type: application/json' -H 'Idempotency-Key: users2-profile-001'
```

\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1

```
curl http://localhost:8080/api/v1/profile-jobs/<job-id>
```

\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1

```
curl http://localhost:8080/api/v1/profile-jobs/<job-id>/result
```

11. Idempotency

[illegible]

```
ALTER TABLE jobs ADD COLUMN IF NOT EXISTS idempotency_key TEXT;  
CREATE UNIQUE INDEX IF NOT EXISTS idx_jobs_idempotency_key ON jobs(idempotency_key) WHERE idempotency_key IS NOT NULL;
```

[illegible]

12. Completeness and Null Analysis

[illegible]

13. Cardinality and Uniqueness

1

14. Numeric Profiling

[illegible]

19. Correlations

[illegible]

20. Relationship Discovery

1

21. Quality Rules and Score

[illegible]

22. How to Analyze a Real Result

[illegible][illegible][illegible][illegible]

|1|

[illegible][illegible]

|1|

[illegible]

|1|

[illegible][illegible]

23. History and Drift

[illegible]

\1

```
{"changed":false,"score_delta":0,"column_changes":[]}
```

\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1\1

```
GET /api/v1/profiles/history?schema=public&table=users2&limit=20
GET /api/v1/profiles/drift?schema=public&table=users2
```

24. Async Job Reliability

A

25. Security

[illegible]

26. Performance Design

[illegible][illegible]

- [illegible]

- Distributed worker claiming is not implemented.
- Metadata migration currently relies on Docker initialization scripts.
- Source DSNs are redacted from responses but remain plaintext in metadata storage.
- Duplicate detection can be expensive.
- Generic name headers can produce PII false positives.
- Performance has not yet been established through a controlled benchmark suite.
- Readiness should eventually verify dependencies rather than only returning an application-level ready state.

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- ## 38. Final Mental Model

[illegible][illegible]

40. References and Further Reading

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[illegible]